

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1795

Tolson 138 kV
October 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1795. The Interconnection Customer proposes a 207.5 MW solar photovoltaic facility interconnecting at the Tolson 138 kV substation in the Virginia area. The request was received on June 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 562 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,350,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Tolson to Orovada 138kV line
Contingency: Atoka to John Svier 345kV
Loading: 112.8% (Rating: 492 MVA)
DFAX: 75.6%
Pre-project loading: 80.9%
2. Facility: Cinder to Sprigg 138kV
Contingency: Neptune to Ragland 345kV
Loading: 106.6% (Rating: 343 MVA)
DFAX: 12.7%
Pre-project loading: 98.9%
3. Facility: Kimper to Sprigg 138kV
Contingency: Ragland to Neptune Substation 345kV
Loading: 106.1% (Rating: 352 MVA)
DFAX: 12.7%
Pre-project loading: 98.6%
4. Facility: Carlton Hills 138/1 kV Transformer Bank
Contingency: Leesburg Picciola to Orovada 138kV L
Loading: 104.4% (Rating: 401 MVA)
DFAX: 70.9%
Pre-project loading: 67.7%
5. Facility: Kimper to Tom Watkins 138kV
Contingency: Catawba to Atoka 345kV
Loading: 103.8% (Rating: 396 MVA)
DFAX: 17.4%
Pre-project loading: 94.6%

6. Facility: Orovada Substation to Carlton Hills 138kV
 Contingency: Orovada to Leesburg Picciola 138kV
 Loading: 103.5% (Rating: 389 MVA)
 DFAX: 70.9%
 Pre-project loading: 65.7%
7. Facility: Daffin Gin Substation to Whiting Ave 138kV
 Contingency: Orovada to Tolson 138kV
 Loading: 103.3% (Rating: 204 MVA)
 DFAX: 11.0%
 Pre-project loading: 92.1%
8. Facility: Whiting Ave to Torrington Terminal 138kV line
 Contingency: Orovada to Tolson 138kV
 Loading: 102.5% (Rating: 215 MVA)
 DFAX: 11.0%
 Pre-project loading: 91.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tolson to Orovada 138kV line	Rebuild line with higher-capacity conductor	\$15,012,000	18% (\$2,755,000)
Cinder to Sprigg 138kV	Reconductor line	\$1,800,000	6% (\$109,000)
Kimper to Sprigg 138kV	Reconductor line	\$6,087,000	6% (\$368,000)
Carlton Hills 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,130,000	22% (\$1,151,000)
Tom Watkins to Kimper 138kV	Reconductor line	\$2,704,000	6% (\$159,000)
Orovada Substation to Carlton Hills 138kV	Reconductor line	\$203,000	22% (\$46,000)
Whiting Ave to Daffin Gin 138kV	Reconductor line	\$2,179,000	8% (\$181,000)
Whiting Ave to Torrington Terminal 138kV line	Reconductor line	\$6,995,000	8% (\$581,000)
	Total allocated to project		\$5,350,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.